WAITING FOR WEALTH:

INHERITANCES AND HOUSEHOLD SAVINGS

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MOTIVATION

- Life-cycle model: foundational model of household finance and economics
- Inheritances account for 25-50% of U.S. household wealth (Feiveson & Sabelhaus (2018), Gale & Scholz (1994). Modigliani (1989), Kotlikoff & Summers (1981))
- Model ignores inheritance receipt
  - Anticipated inheritances reduce savings
  - The size, probability, and timing of inheritances all matter
  - Most retirement planning software asks about inheritance expectations

BIG QUESTIONS:

1. What features of reality does the model get wrong without inheritance?
   - Miss positive tail risk in income
   - Overstate savings motive
   - Wealth inequality: underestimated at middle age, overstimated for old

2. Implications of inheritances on structural parameters?
   - Ignoring inheritances $\Rightarrow$ overestimate discount factor β

3. Do households have well calibrated expectations about inheritances?
   - Not today

METHOD

- We extend standard life-cycle savings model to include inheritances
- Households are endowed with a parent
- Parents’ wealth evolves stochastically
- Households receive an inheritance when their parent dies
- Calibrate inheritance processes using SCF, PSID, HRS, SSA, …

RELATED LITERATURE

1. **Empirical literature on inheritance and wealth persistence:** Kotlikoff & Summers (1981, JPE), Modigliani (1988, JEP), Gale & Scholz (1994, JEP), Charles & Hurst (2003, JPE), Gokhale, Kotlikoff, Sefton, & Weale (2001, JPubE), Piketty (2011, QJE), Boserup, Kopczuk & Kreiner (2016, AER), Elinder, Erixson, Waldenstrom (2018, JPubE), …

   This paper: Endogenous responses to inheritance expectations and receipt

2. **Life-cycle models with bequests (parent's side):** De Nardi (2004, REStud), De Nardi, French & Jones (2015, JPE), Lockwood (2018, AER), …

   This paper: Models child's side

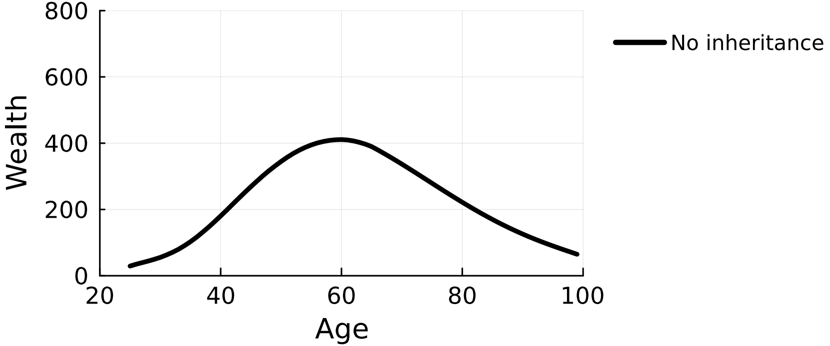
3. **Anticipated inheritances and household behavior:** Doorley & Pestel (2020, OBES), Coile & Weisbenner (2010, REStat), Kindermann, Mayr & Sachs (2020, JPubE), Aliprant, Carrol, & Young (2025), Asher (2026), …

   This paper: Structural model capturing probability, timing, and size jointly

4. KMS (2020, JPubE) Closest; focus on estate taxes and labor response.

STYLIZED EXAMPLE

HOW DO LARGE INHERITANCES AFFECT AGGREGATE WEALTH?



- No inheritance: standard hump-shaped wealth profile
- Certain inheritance: households anticipate inheritances and adjust savings
- 50% inheritance: Probability, size, and timing of inheritance all matter
  - Uncertain timing

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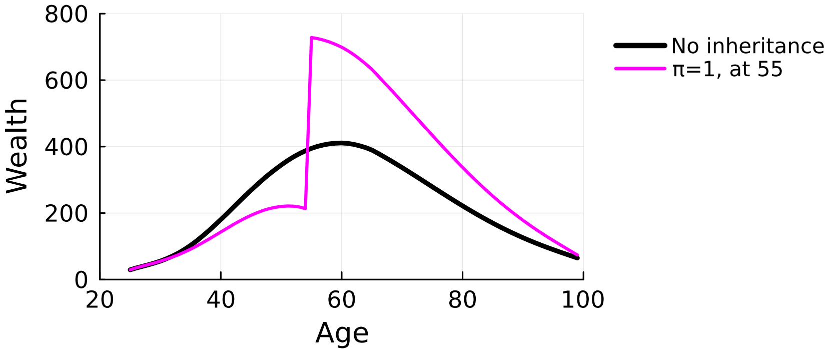

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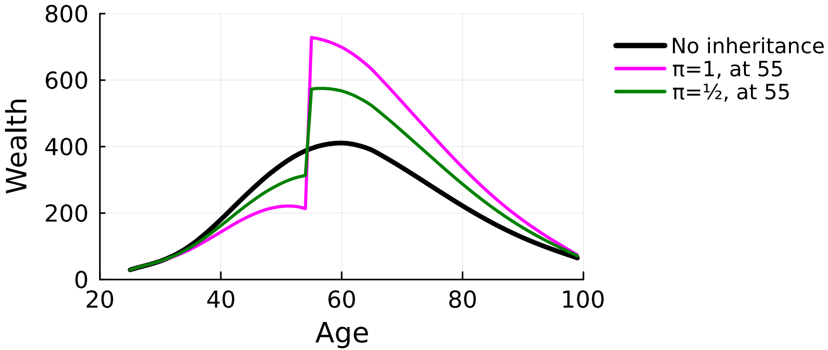

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A SIMPLE MODEL OF INHERITANCE RECEIPT

STANDARD LIFE-CYCLE MODEL

- Households live from age a 25 to 100, retire at 65
- CRRA preferences, warm-glow bequest motive
- Income: deterministic life-cycle + persistent (v) + transitory (ε) shocks
- Save in a risk-free asset; no borrowing
- Budget constraint:

$$\frac{x'}{1+r} = x - c + y(v', \varepsilon')$$

‣ Full model

ADDING INHERITANCES

- New state: parental status p — encodes survival and wealth
  - Dead (absorbing)
  - Alive at wealth quintile $w_1, \ldots, w_5$ (per child)
  - Age-varying mortality
  - Age-varying Markov chain for parental wealth: $w' \mid w, a \sim P_{ww'}(a)$
  - Budget constraint:

$$\frac{x'}{1+r} = x - c + y(v', \varepsilon') + \mathbf{1}_{\text{death}} \cdot (1 - \tau(w))\, w$$

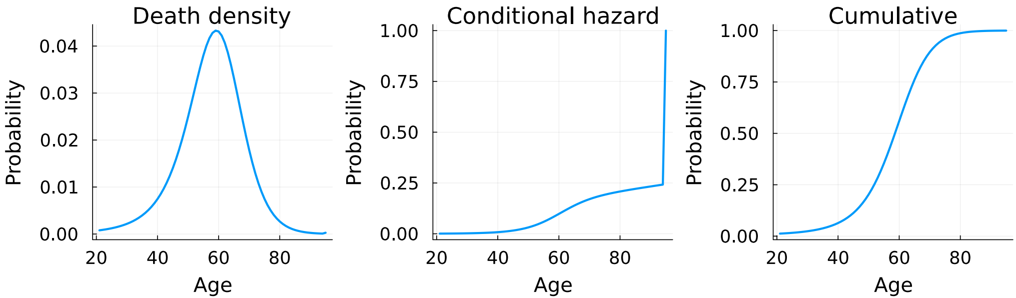
- $\tau(w)$ = taxes and fees; varies by quintile
- Three sources of uncertainty: probability, size, and timing of inheritance

▸ Full model

CALIBRATING THE PARENTAL PROCESSES

PARENTAL MORTALITY

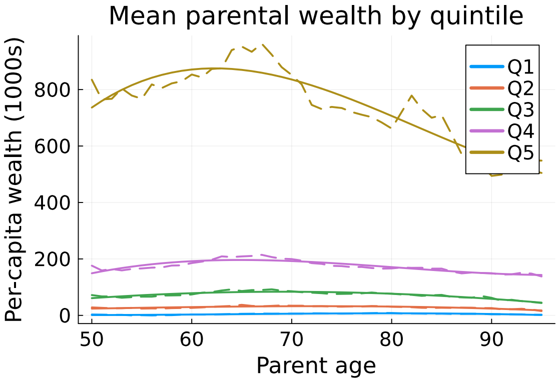
- Last parent death probabilities derived from SCF



Inheritance receipt at 50–70 ⇒ expectations matter — but ignored by most models

PARENTAL WEALTH QUINTILES (HRS) - PER LIBEROS (CHILD)

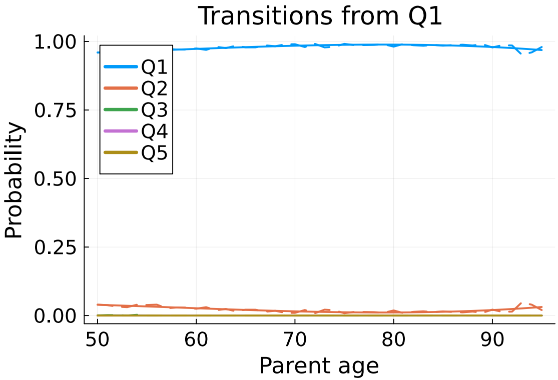
- Age-dependent quintile grid: $\mathcal{W}(a) = \{w_1(a), \ldots, w_5(a)\}$



Top two quintiles: \$800k and \$200k ⇒ relevant not only for "top 1 percent"

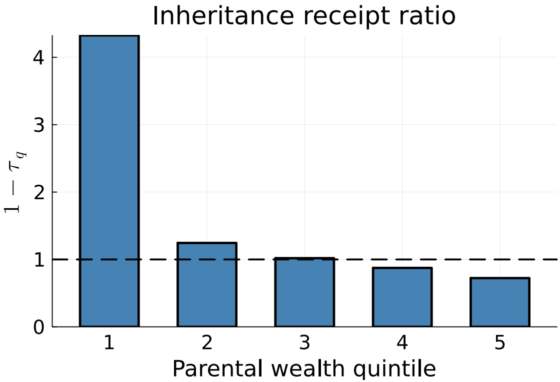
PARENTAL WEALTH TRANSITIONS (HRS)

- Markov chain with age-varying transition matrices: $\Pr(w' \mid w, a)$



INHERITANCE SLIPPAGE

- Slippage $\tau(w)$: fraction of parental wealth lost to taxes, fees, etc.
- Estimated from HRS exit interviews

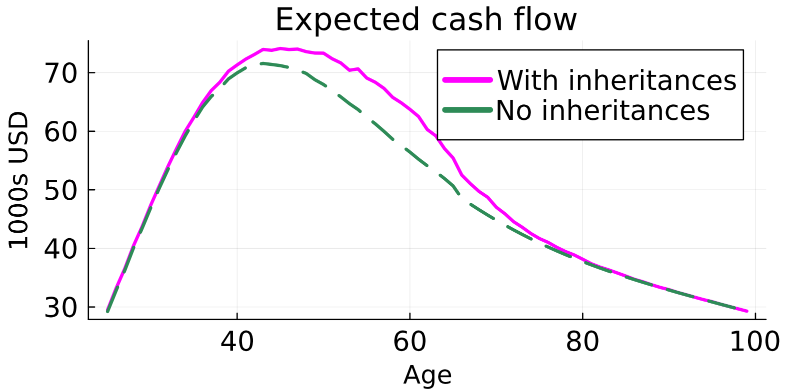


Poorest: bequests from life insurance, not wealth. Richest: ~30% lost to slippage.

‣ Net inheritance by age

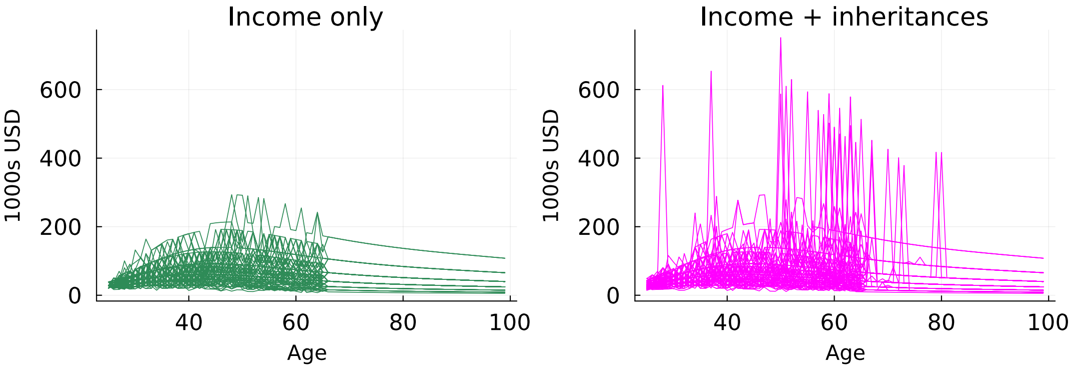
EFFECT OF INHERITANCES ON ~~INCOME~~ CASH FLOW

EXPECTED ~~INCOME~~ CASH FLOW: WITH VS. WITHOUT INHERITANCES



- Inheritances add a mid-life income bump (ages 50–70)

INHERITANCES GENERATES LARGE ~~INCOME~~ CASH FLOW SPIKES



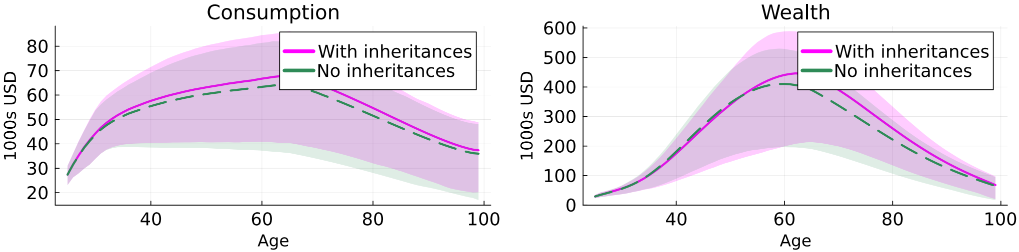
- Simulated income paths for 100 households
- Inheritances create large positive income spikes ⇒ positive tail risk

CALIBRATION

- We have now calibrated all inheritance processes
  - Age-dependent parental wealth quintiles
  - Age-dependent parental wealth transitions
  - Age-dependent parental mortality
  - $\Rightarrow$ Stochastic process determining inheritance receipt
- Every else is standard $\Rightarrow$ take values from literature

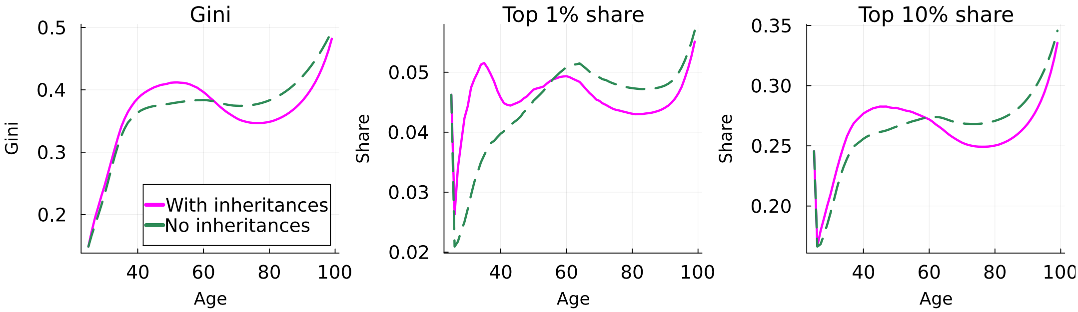
AGGREGATE EFFECTS OF INHERITANCES

LIFE-CYCLE PROFILES: WITH VS. WITHOUT INHERITANCES



- Inheritances increase consumption at all ages
- Inheritances increase wealth for older households
- Inheritances decrease savings (out of income) for younger households

WEALTH INEQUALITY: WITH VS. WITHOUT INHERITANCES



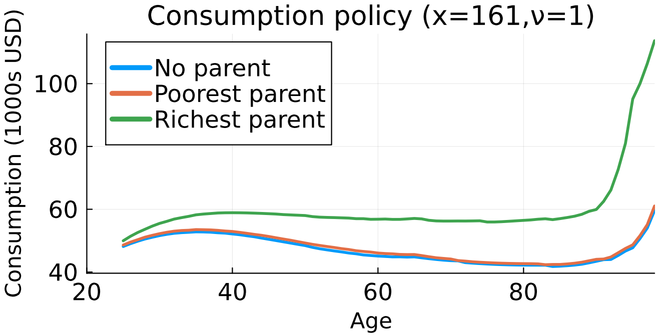
- Inheritances ⇒ increase wealth inequality among young but lower it for old
- Partially resolves old debate on inheritances and wealth inequality (Modigliani (1988))

‣ Percentile ratios

CONSUMPTION GRADIENT BY PARENTAL WEALTH

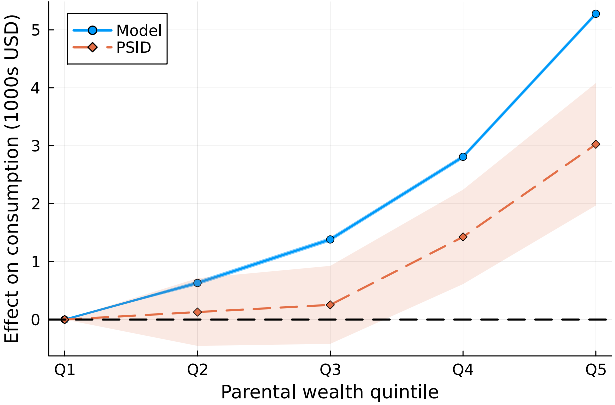
CONSUMPTION POLICY BY PARENTAL STATE

- Expect future inheritance: save less and consume more
- Poor parent ≈ dead parent
- Wealthier parents, all else equal, increases consumption by $5-20k per year



CONSUMPTION GRADIENT: REGRESSION

- Regress consumption on parental wealth quintile dummies
  - Controls: own wealth, income, and demographics.
    - Regression details
- Q1 = reference group
- Model matches PSID gradient: richer parents $\Rightarrow$ higher consumption



THE VALUE OF INHERITANCES

WILLINGNESS TO PAY FOR INHERITANCE RECEIPT TODAY

- How much would a household pay to receive the inheritance today?

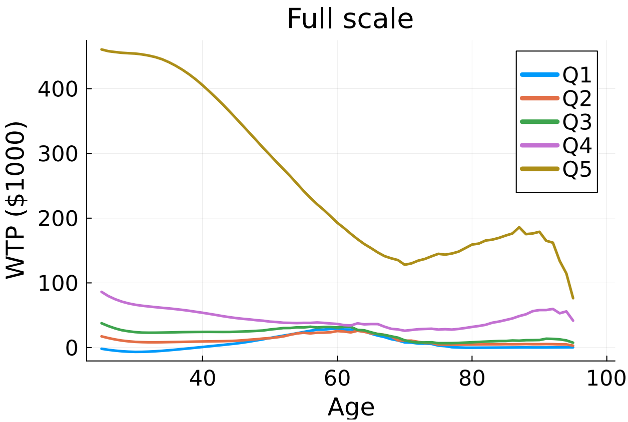
$$V(x, v, p, a) = V(x - \text{WTP} + w(1 - \tau), v, \text{dead}, a)$$

- WTP $>$ 0: household benefits from early death
- WTP $<$ 0: option value of living parent exceeds inheritance.
- Use model to understand which frictions and market incompleteness drive the WTP

WTP FROM SIMULATED PANEL

- Young households with rich parents: large positive WTP (want inheritance now)
- WTP generally decreasing in age
    1. Need money less
    2. Shorter time to inheritance
    3. Avoid parental dissaving

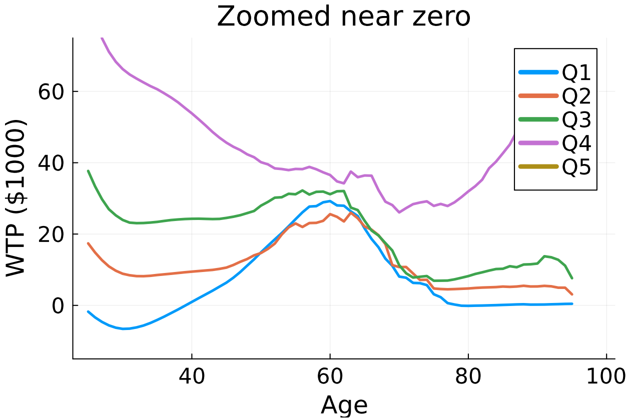
▸ Net inheritance by age



WTP FROM SIMULATED PANEL

- Q1–Q3 parents: near-zero or negative WTP
- WTP < 0: option value of living parent exceeds the inheritance
- Age 60: WTP > 0 since its peak parental wealth

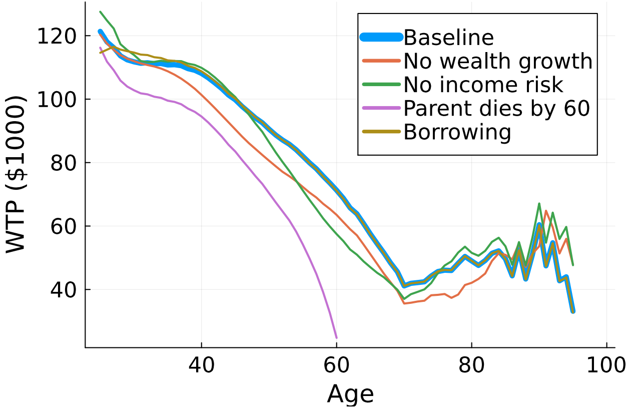
▸ Net inheritance by age



DECOMPOSING THE WTP

WTP under different models:

- Fixed parental wealth quintile
- No income risk
- Parents die by 60
- Allow borrowing
- Uncertainty resolution important
  - Timing and amount
- Borrowing not important



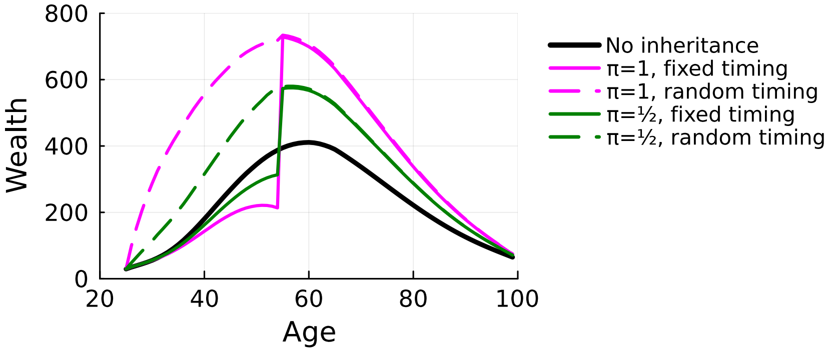
▸ By parental wealth

(PRELIMINARY) CONCLUSION

- Background: Model omits inheritances, source of 25-50% of private wealth
- Question: Are inheritances important?
- Method: Extend life-cycle model to include inheritances
- Results:
  - Calibrated Inheritances $\Rightarrow$ extremely large positive income shocks
  - Omitting inheritances $\Rightarrow$ overestimate discount factor β
  - Inheritances increase consumption more than wealth
- Next steps:
  - Event study of wealth and consumption after inheritances
  - Are inheritance expectations well calibrated?

UNCERTAIN TIMING OF INHERITANCE

Adding random timing (dashed): inheritance may arrive at any age before 55.



‣ Return

STANDARD LIFE-CYCLE MODEL

- Households live from age 25 to 100, retire at 65
- Choose consumption c to maximize lifetime utility:

$$V(x, v, a) = \max_{c} \; u(c) + \beta \, p_a \, \mathbb{E}\left[V(x', v', a+1)\right] + \beta(1 - p_a)\, \phi(x')$$

- CRRA flow utility $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$, warm-glow bequest $\phi(x) = \psi_0 \frac{(x+\psi_1)^{1-\gamma}}{1-\gamma}$
- Budget constraint:

$$x' = (1+r)(x-c) + y(v', \varepsilon')$$

- Income: persistent (v) + transitory (ε) shocks, hump-shaped deterministic profile

▸ Return

ADDING INHERITANCES

- Parental status $p \in \{\text{dead}, \; w_1, \ldots, w_5\}$
- Extended value function:

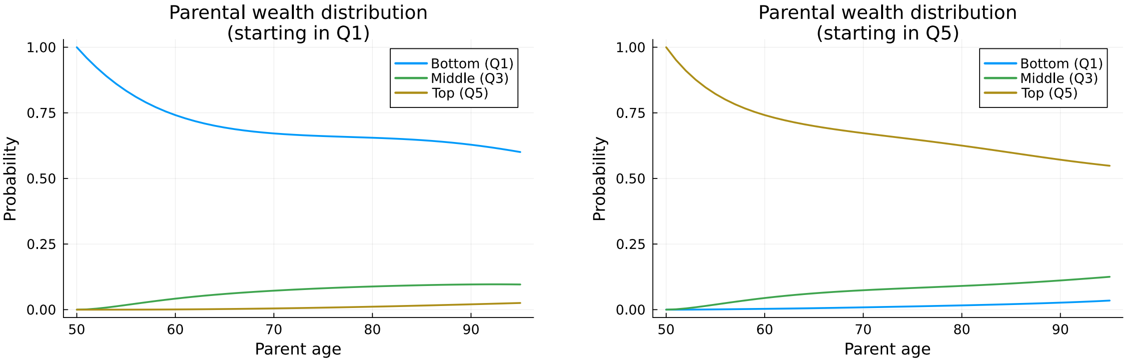
$$V(x, v, p, a) = \max_c \; u(c) + \beta \, p_a \, \mathbb{E}\left[V(x', v', p', a+1)\right] + \beta (1 - p_a) \, \phi(x')$$

- Budget constraint: $x' = (1 + r)(x - c) + y(v', \varepsilon') + \mathbf{1}_{\text{death}} \cdot (1 - \tau(w)) \, w$
- Parental transitions:
  - Survival: age-varying SSA death probabilities
  - Wealth: quintile-to-quintile transitions from PSID
  - Slippage $\tau(w)$: taxes, fees, other heirs (varies by quintile)
- Once parent dies (p = dead), model collapses to the standard life-cycle problem

▸ Return

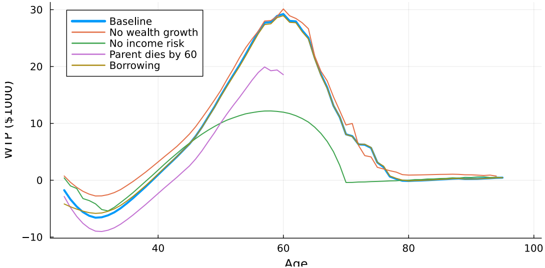
PARENTAL WEALTH QUINTILE DYNAMICS

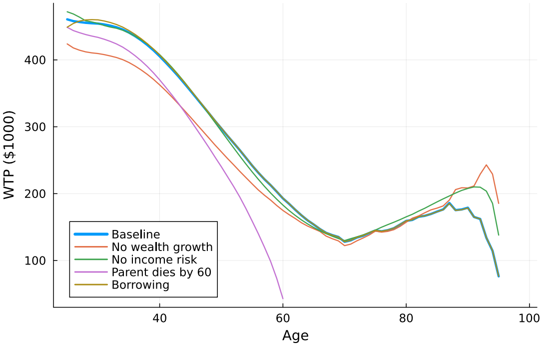
Probability of being in each quintile by age, given initial parental wealth at age 25.



‣ Return

WTP DECOMPOSITION BY PARENTAL WEALTH





Left: poorest parents. Right: richest parents.

- Return

REGRESSION SPECIFICATION

$$c_{it} = \alpha + \sum_{q=2}^{5} \beta_q \mathbf{1}[Q_{it} = q] + \gamma' X_{it} + \varepsilon_{it}$$

Q_{it}: parental wealth quintile (Q1 = reference)

- **PSID:** X_{it} = own wealth, quadratic in income, quadratic in age, education, race, marital status, homeownership, survey year. Family weights, clustered SEs.
- **Model:** X_{it} = own wealth, income, persistent income state indicators, quadratic in age.

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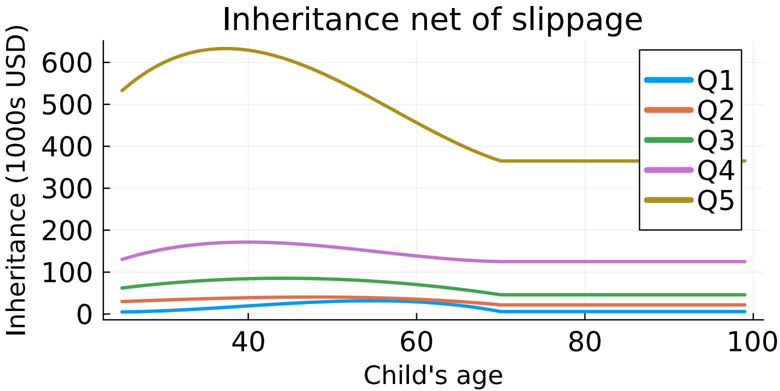
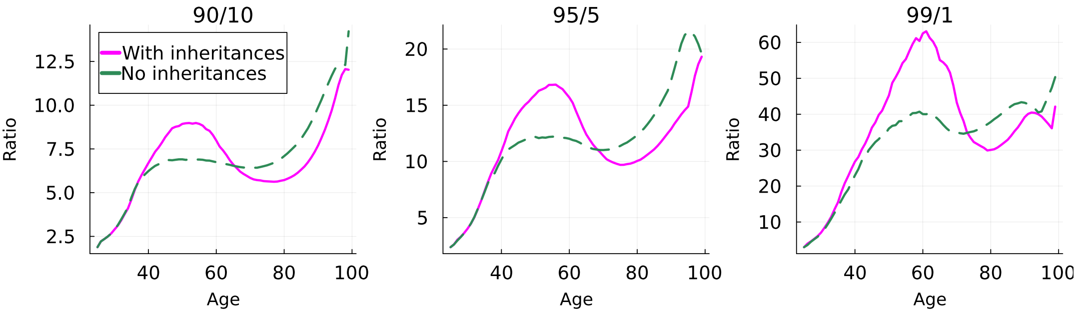
INHERITANCE NET OF SLIPPAGE BY CHILD'S AGE



Amount received $(1 - \tau(w))\, w(a)$ if parent dies at each age, by parental wealth quintile.

‣ Slippage ‣ WTP

WEALTH INEQUALITY – SPREAD



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